CMSC 125 Course Guide

Operating Systems

University of the Philippines

1st Semester AY 2024-2025

Contents

1 Course Description

Course Number	CMSC 125
Course Title	**Operating Systems**
Description	Processor management, memory management, file and disk management, resource management, concurrent processes, networks and distributed systems.
Prerequisite	CMSC 123 or COI

2 Course Introduction

Welcome to CMSC 125! The aim of this course is to introduce students to operating systems concepts.

3 Course Objectives

At the end of the course, you should be able to:

- describe the purpose and functions of an operating system;
- design and implement programs to perform simple operating system tasks; and
- evaluate the performance of different operating system algorithms.

4 Course Prerequisite

This course will require that you have taken CMSC 123, Data Structures.

5 Course Materials and Schedule

- You will be provided with **video lectures** and a PDF version of the set of **presentation slides** used in the videos as well as a set of **labs** for you to do, mostly coding. These will be made available through the course website or Google Classroom. No coursepack will be provided.
- Below is the timetable summarizing the topics that we will cover for the semester. We advise that during each week you spend one to two hours viewing the videos of the lectures. Don't forget to read the textbook!

Wk	Scheduled Dates	Actual Dates	Topics	Expected Learning Outcomes	Learning Resources	Labs	F2F/Async Activities
1	**Aug 19-23**		Introduction to the Course			Tutorial01: Installing Ubuntu Desktop 20.04 on VirtualBox	*F2F*: Discussion/Quizzes/Lab *Async*: Readings/Videos/Consultation via Slack
2	**Aug 26-30**	*Aug 26-30*	-Introduction to OS -The Process Abstraction -Process API Example	-Describe the main services provided by an operating system in the context of virtualization, concurrency, and persistence -Differentiate a program from a process -Describe the different states a process can be in -Differentiate CPU-bound process and I/O-bound processes -Explain how a process transitions from one state to another -Explain how the fork(), wait(), and exec() system calls work -Describe other process control functions	OSTEP Ch. 2 OSTEP Ch. 4 OSTEP Ch. 5 YouTube Videos	Lab01: Advanced Linux Commands	*F2F:* Discussion/Quizzes/Lab *Async:* Readings/Videos/Consultation via Slack

Wk	Scheduled Dates	Actual Dates	Topics	Expected Learning Outcomes	Learning Resources	Labs	F2F/Async Activities
3	**Sep 2-6**	*Sep 2-6* *Sep 9-13*	Limited Direct Execution Scheduling Multi-Level Feedback Queue Fair-Share Scheduling Multiprocessor Scheduling	Illustrate the Limited Direct Execution Protocol Describe how the trap and return-from-trap instructions work Differentiate user mode from kernel mode Describe the approaches on how the kernel takes back control when switching processes Explain how the timer interrupt is used for the kernel to regain control Evaluate various CPU scheduling policies based on certain metrics given some assumptions on the input processes Describe how MLFQ works Describe the different approaches to proportional share scheduling Describe the challenges, and solutions, in multiprocessor scheduling	OSTEP Ch. 6 OSTEP Ch. 7 OSTEP Ch. 8 OSTEP Ch. 9 OSTEP Ch. 10 YouTube Videos	Lab02: BASH Shell Scripting	*F2F:* Discussion/Quizzes/Lab *Async:* Readings/Videos/Consultation via Slack

Wk	Scheduled Dates	Actual Dates	Topics	Expected Learning Outcomes	Learning Resources	Labs	F2F/Async Activities
4	**Sep 9-13**	*Sep 16-20*	Address Spaces Memory API Address Translation	Explain what the address space of a process is Illustrate the address space of a process Explain the function and behaviour of each section of a process' address space: code, data, stack, heap Describe the goals for virtualizing memory Explain how the malloc(), free(), sbrk(), and other functions are used in C programs Explain the importance and mechanisms of address translation: virtual address to physical address Illustrate Limited Direct Execution with Dynamic Relocation at boot time and at run time Illustrate internal fragmentation	OSTEP Ch. 13 OSTEP Ch. 14 OSTEP Ch. 15 YouTube Videos	Lab03: Process API in Linux	*F2F:* Discussion/Quizzes/Lab *Async:* Readings/Videos/Consultation via Slack

Wk	Scheduled Dates	Actual Dates	Topics	Expected Learning Outcomes	Learning Resources	Labs	F2F/Async Activities
5	**Sep 16-20**	*Sep 23-27* *Sep 30-Oct 4* *Oct 7-11*	Segmentation Free Space Management Paging TLB Swapping Mechanisms Advanced Page Tables	Illustrate how segmentation works Explain how the hardware determines the segment and offset within a segment given a memory address Illustrate external fragmentation Describe a basic algorithm for implementing malloc() Illustrate how paging works Illustrate how the TLB helps in improving paging performance Illustrate how paged-segmentation works Illustrate how multi-level paging works	OSTEP Ch. 16 OSTEP Ch. 17 OSTEP Ch. 18 OSTEP Ch. 19 OSTEP Ch. 20 YouTube Videos	Lab04: CPU Scheduling Simulations	*F2F:* Discussion/Quizzes/Lab *Async:* Readings/Videos/Consultation via Slack
6	**Sep 23-27**		Policies and Mechanisms for Memory (reading assignment)	Explain the role and implementation of swap space Illustrate what happens during a page fault Explain what thrashing is	OSTEP Ch. 21 OSTEP Ch. 22 OSTEP Ch. 23 YouTube Videos	Lab05: Understanding the PC Boot Process	*F2F:* Discussion/Quizzes/Lab *Async:* Readings/Videos/Consultation via Slack

Wk	Scheduled Dates	Actual Dates	Topics	Expected Learning Outcomes	Learning Resources	Labs	F2F/Async Activities
7	**Sep 30-Oct 4**		(Review and Exam Week)			Lab06: Virtual Memory Simulation	
EXAM 1 (Oct 3, Thu, 7PM-9PM)							
8	**Oct 7-11**	*Nov 4-8*	Concurrency Thread API Locks	Explain what a thread is and its advantages Differentiate a process from a thread Explain terms related to concurrency: race condition, critical section, mutual exclusion, atomicity Write multithreaded programs using the PThreads API Describe the lock abstraction Describe and evaluate different lock implementations Describe how basic data structures are implemented to support concurrent access	OSTEP Ch. 26 OSTEP Ch. 27 OSTEP Ch. 28 OSTEP Ch. 29 YouTube Videos	Lab07: Unix System V Shared Memory API	*F2F:* Discussion/Quizzes/Lab *Async:* Readings/Videos/Consultation via Slack
Oct 14-19: READING BREAK							

Wk	**Scheduled Dates**	**Actual Dates**	**Topics**	**Expected Learning Outcomes**	**Learning Resources**	**Labs**	**F2F/Async Activities**
9	**Oct 21-24**	*Nov 11-15*	Lock-based concurrent data structures Condition Variables Semaphores	Articulate when and how to use condition variables in concurrent programs Articulate when and how to use semaphores in concurrent programs Describe and evaluate solutions to the Bounded-Buffer Problem Articulate when and how to use reader-writer locks in concurrent programs Describe and evaluate solutions to the Dining Philosophers Problem Implement semaphores using locks and condition variables Describe some non-deadlock concurrency bugs Explain why deadlocks occur Describe the approaches on how to prevent and avoid deadlocks as well as detect and recover from it Describe how event-based concurrency works Explain how select() works Explain what asynchronous I/O is	OSTEP Ch. 30 OSTEP Ch. 31 OSTEP Ch. 32 OSTEP Ch. 33 YouTube Videos	Lab08: Programming with PThreads	*F2F:* Discussion/Quizzes/Lab *Async:* Readings/Videos/Consultation via Slack

Wk	Scheduled Dates	Actual Dates	Topics	Expected Learning Outcomes	Learning Resources	Labs	F2F/Async Activities
10	**Oct 28-Nov 1**	*Nov 18-22*	Common Concurrency Problems Event-based Concurrency I/O Devices HDD	Describe modern computer system architecture Describe a basic I/O device and its interface Differentiate programmed I/O from interrupt-driven I/O Explain how Direct Memory Access (DMA) works Describe approaches on how the CPU interacts with the hardware: I/O instructions and memory-mapped I/O Describe what a device driver is and how it interacts with the kernel Explain how a simple IDE Disk driver works Explain the interface and geometry of hard disk drives Explain the factors that contribute to disk performance Calculate the I/O time and rate of I/O of a disk given a disk specification and different workloads Describe various disk scheduling algorithms and their properties	OSTEP Ch. 36 OSTEP Ch. 37 OSTEP Ch. 38 YouTube Videos	Lab09: Thread Synchronization in PThreads	*F2F:* Discussion/Quizzes/Lab *Async:* Readings/Videos/Consultation via Slack

Wk	Scheduled Dates	Actual Dates	Topics	Expected Learning Outcomes	Learning Resources	Labs	F2F/Async Activities
11	**Nov 4-8**	*Nov 25-30* *Dec 2-6*	Files and Directories File System Implementation	Explain the file and directory abstractions as well as associated terminologies Describe and use the File System API: creating, reading and writing, renaming, mounting, etc. Describe a simple File System Implementation Explain the importance of caching and buffering in File Systems Describe how the Fast File System works	OSTEP Ch. 39 OSTEP Ch. 40 OSTEP Ch. 41 OSTEP Ch. 42 YouTube Videos	Lab10a: [ICS-OS] Building and Booting ICS-OS Lab10b: [ICS-OS] Command Line Interface, System Calls, and System Utilities	*F2F:* Discussion/Quizzes/Lab *Async:* Readings/Videos/Consultation via Slack
12	**Nov 11-15**		(buffer week)			Lab11: [ICS-OS] Environment Variables, Processes, and Threads	*F2F:* Discussion/Quizzes/Lab *Async:* Readings/Videos/Consultation via Slack
13	**Nov 18-22**		(buffer week)			Lab12: [ICS-OS] Implementing Lottery Scheduling in ICS-OS	
14	**Nov 25-30**		(Review and Exam Week)				
EXAM 2							

Wk	Scheduled Dates	Actual Dates	Topics	Expected Learning Outcomes	Learning Resources	Labs	F2F/Async Activities
15	**Dec 2-6**						
16	**Dec 9-12**						

6 References

- **[OSTEP]** Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau. 2018. Operating Systems: Three Easy Pieces. CreateSpace Independent Publishing Platform, North Charleston, SC, USA. (*http://pages.cs.wisc.edu/~remzi/OSTEP/*)
- **[OSBOOK10]** Operating System Concepts, 10th Edition by Abraham Silberschatz, Peter B. Galvin, Greg Gagne (*https://www.os-book.com/OS10/*)
- **[ICS-OS]** J. A. C. Hermocilla. Ics-os: A kernel programming approach to teaching operating system concepts. Philippine Information Technology Journal, 2(2):25–30, 2009. (*https://github.com/srg-ics-uplb/ics-os*).

7 Grading

Your **final standing** will be based on the following requirements. No Final Exam.:

Component	**Weight**
Two Lecture Exams	40%
Lecture Quizzes/Homework/Problem Set	10%
ICS-OS Project	10%
Laboratory	40%
Total	**100%**

We shall be using the following **grading scale** to transmute your final standing to a numerical grade:

Range (Lower)	**Range (Upper)**	**Grade**
95.00	100.00	1.00
90.00	94.99	**1.25**
85.00	89.99	**1.50**
80.00	84.99	**1.75**
75.00	79.99	**2.00**
70.00	74.99	**2.25**
65.00	69.99	**2.50**
60.00	64.99	**2.75**
55.00	59.99	**3.00**
0.00	54.99	**5.00**

8 On Course Requirements

8.1 Attendance

- Attendance is required. Rules on excessive absences will apply. A student will be marked late when he/she arrives 20 minutes after the class has started. Three lates are equivalent to 1 unexcused absence.

8.2 Exams

- Lecture Exams will be in person which may be scheduled outside of class hours.

8.3 Labs

- You will need an Ubuntu Desktop 20.04 (64-bit, virtual or physical). We will use Docker for some labs including ICS-OS Labs. Delivery and submission of lab exercises will be done through Github Classroom. Links to the labs will be posted in Slack or Google Classroom.

8.4 Other Activities

- We will use a Slack workspace for asynchronous discussions and synchronous sessions.

9 On Academic Integrity

Intellectual honesty is one of the most important values which this university upholds. Therefore, as a student of the University of the Philippines, it is expected that any work you submit for this course is your own. This means that cheating in exercises, plagiarism and other forms of intellectual dishonesty will not be tolerated. It is said that each one of us has his own brilliant idea and therefore you should not miss the chance of discovering yours.

If you are caught cheating in this course, the policy below will be followed:

- for the first offense, you will receive a score of zero for quiz/exam/exercise; and
- for the second offense, you will receive a final grade of 5.0 for the course.

10 Teaching Staff

The faculty members handling the course for this semester are the following:

- Joseph Anthony C. Hermocilla (jchermocilla@up.edu.ph)
- Aldrin Joseph J. Hao (ajhao@up.edu.ph)
- Leonard Paul Garchitorena (lagarchitorena@up.edu.ph)
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Feel free to email or message them in Slack if you have concerns about CMSC 125 this semester.

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